CS511 – Web Engineering

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1. What is the difference between inline and block-level HTML elements?
2. How would you make a simple PHP class serializable?
3. What are the two types of views supported by Laravel?
4. Name two development stacks that utilize Node.js as the backend server.
5. List at least three design patterns commonly used in web application development.
6. Write JavaScript code for functions "Func1", "Func2", and "Func3". In "Func3", call "Func2" with "Func1" as a callback. Then, within "Func2", call "Func1" using the callback mechanism.
7. Write PHP code to:
    - a) Set a cookie named "Theme" on the client-side.
    - b) Check if the cookie named "Theme" exists.
8. Using SimpleXML in PHP, write code to access the "title" attribute of the first "painting" element in an XML file named "art.xml".
9. List at least three features of a Node.js web server.
10. Describe the design pattern that allows for easy switching of databases (e.g., from MySQL to SQL Server) with minimal code changes. Explain the reasoning behind this pattern.
11. Differentiate between the "==" (Equal-To) and "===" (Identical) operators in PHP.
12. Discuss the details of two different categories of Top-Level Domains (TLDs).
13. The following JavaScript function displays the value of an undefined variable ("y"). Modify the function using exception handling to catch and display an error message to the console.

    JavaScript
    ```
    function myFunction() {
      console.log(y);
    }
    ```
14. Name three CSS preprocessors.
15. How many breakpoints are defined in the Bootstrap grid system?
16. What does "RD" stand for?
17. What font family does Arial belong to?
18. What does CMTS stand for?
19. Briefly describe Web 1.0, Web 2.0, and Web 3.0.
20. What is the purpose of console.log() in JavaScript?
21. Explain the difference between dynamic and static websites.
22. Define CSS selectors.
23. Describe a query in the context of web development.
24. What is the z-index property in CSS?
25. What is the result of Math.abs(-5)?
26. How does Bootstrap utilize a grid layout for columns?
27. What are block-level elements?
28. What are inline elements?
29. In CSS, what does "3em" typically refer to?
30. If no URL is specified in an HTML document, which file will be served by the server by default?
31. List five semantic HTML tags.
32. Given a sample HTML code, write CSS code to change the color of <p> and <h1> elements to red.
33. Write a JavaScript function to sum two integers.
34. How is a table created in Laravel using Eloquent ORM?
35. Provide a basic PHP class definition for a "Student" class.
36. Name two development stacks that utilize Node.js as the backend server.
37. List at least three common threat vectors in web applications.
38. Write PHP code to check if a cookie named "user_name" is set.
39. Create a basic PHP script that attempts to connect to a database (you can choose a database system like MySQL).
40. Write a PHP script that starts a session and checks if a user is logged in.
41. List the common HTTP methods used in web services.

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Answers:

1. What is the difference between inline and block-level HTML elements?

- **Inline Elements:** Take up only as much width as necessary. They don't start on a new line. Example: <a>, , <img>.
- **Block-level Elements:** Take up the full width of their container and start on a new line. Example: , <p>, <h1>.

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2. How would you make a simple PHP class serializable?

Use the Serializable interface and implement its serialize() and unserialize() methods.

```
class MyClass implements Serializable {
   private $data;

   public function __construct($data) {
      $this->data = $data;
   }

   public function serialize() {
      return serialize($this->data);
   }

   public function unserialize($data) {
      $this->data = unserialize($data);
   }
}
```

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3. What are the two types of views supported by Laravel?

1. **Blade Views:** A templating engine provided by Laravel.
2. **Raw Views:** Basic HTML views without using Blade features.

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4. Name two development stacks that utilize Node.js as the backend server.

1. **MEAN Stack:** MongoDB, Express.js, Angular, Node.js.
2. **MERN Stack:** MongoDB, Express.js, React, Node.js.

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5. List at least three design patterns commonly used in web application development.

1. **MVC (Model-View-Controller)**
2. **Singleton**
3. **Observer**

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6. Write JavaScript code for functions "Func1", "Func2", and "Func3".

```
function Func1() {
   console.log("Func1 called");
}

function Func2(callback) {
   console.log("Func2 called");
   callback();
}

function Func3() {
   Func2(Func1);
}

Func3();
```

---

7. Write PHP code to:

a) Set a cookie named "Theme" on the client-side:

```
setcookie("Theme", "dark", time() + 3600, "/");
```

b) Check if the cookie named "Theme" exists:

```
if (isset($_COOKIE["Theme"])) {
   echo "Cookie 'Theme' is set!";
} else {
   echo "Cookie 'Theme' is not set.";
}
```

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8. Using SimpleXML in PHP, access the "title" attribute of the first "painting" element in "art.xml".

```
$xml = simplexml_load_file("art.xml");
echo $xml->painting[0]['title'];
```

9. List at least three features of a Node.js web server.

1. **Asynchronous and non-blocking.**
2. **Event-driven architecture.**
3. **Supports a large number of simultaneous connections.**

10. Describe the design pattern for easy database switching.

- **Pattern:** Repository Pattern.
- **Reason:** Separates database logic from the application logic, allowing you to change the database system (e.g., MySQL to SQL Server) with minimal changes.

11. Differentiate between "==" and "===" in PHP.

- **"==":** Checks if values are equal, ignoring data type.
- **"===":** Checks if values and data types are equal.

12. Discuss two categories of Top-Level Domains (TLDs).

1. **Generic TLDs (gTLDs):** Commonly used, like .com, .org, .net.
2. **Country-Code TLDs (ccTLDs):** Specific to countries, like .uk, .in.

13. Modify the JavaScript function with exception handling.

```
function myFunction() {
  try {
    console.log(y);
  } catch (error) {
    console.error("Error: " + error.message);
  }
}
```

14. Name three CSS preprocessors.

1. **SASS**
2. **LESS**
3. **Stylus**

15. How many breakpoints are defined in the Bootstrap grid system?

Five breakpoints: xs, sm, md, lg, xl.

16. What does "RD" stand for?

- **RD:** Responsive Design.

17. What font family does Arial belong to?

- **Sans-serif** family.

18. What does CMTS stand for?

- **CMTS:** Cable Modem Termination System.

19. Briefly describe Web 1.0, Web 2.0, and Web 3.0.

- **Web 1.0:** Static web pages.
- **Web 2.0:** Interactive and dynamic (e.g., social media).
- **Web 3.0:** Decentralized and AI-powered.

20. What is the purpose of console.log() in JavaScript?

To display messages or debug information in the browser console.

21. Explain the difference between dynamic and static websites.

- **Static:** Fixed content, no server-side processing.
- **Dynamic:** Content changes based on user input or server-side processing.

22. Define CSS selectors.

CSS selectors are patterns used to select and style HTML elements.

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23. Describe a query in web development.

A request to fetch or manipulate data, often sent to a database.

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24. What is the z-index property in CSS?

Defines the stack order of elements (higher values appear in front).

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25. What is the result of Math.abs(-5)?

- **Result:** 5.

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26. How does Bootstrap utilize a grid layout for columns?

Bootstrap divides the grid into 12 columns and uses classes like col- for layout.

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27. What are block-level elements?

Elements that take up the full width and start on a new line.

---

28. What are inline elements?

Elements that only take as much width as needed and don't start on a new line.

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29. In CSS, what does "3em" refer to?

Three times the size of the parent element's font size.

---

30. If no URL is specified, which file will the server serve by default?

The **index.html** file.

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31. List five semantic HTML tags.

1. <header>
2. <footer>
3. <article>
4. <section>
5. <main>

---

32. Write CSS code to change <p> and <h1> colors to red.

```
p, h1 {
  color: red;
}
```

---

33. Write a JavaScript function to sum two integers.

```
function sum(a, b) {
  return a + b;
}
```

---

34. How is a table created in Laravel using Eloquent ORM?

Use a migration:

php artisan make:migration create_table_name

---

35. Provide a basic PHP class definition for a "Student" class.

```
class Student {
  public $name;

  public function __construct($name) {
    $this->name = $name;
  }
}
```

---

36. Name two development stacks that utilize Node.js as the backend server.

1. **MEAN Stack**
2. **MERN Stack**

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37. List three common threat vectors in web applications.

1. SQL Injection
2. Cross-Site Scripting (XSS)
3. Cross-Site Request Forgery (CSRF)

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38. Write PHP code to check if a cookie named "user_name" is set.

```
if (isset($_COOKIE["user_name"])) {
   echo "Cookie 'user_name' is set.";
} else {
   echo "Cookie 'user_name' is not set.";
}
```

---

39. Create a PHP script to connect to MySQL.

```
$conn = new mysqli("localhost", "username", "password", "database");
if ($conn->connect_error) {
   die("Connection failed: " . $conn->connect_error);
}
echo "Connected successfully";
```

---

40. Write a PHP script to start a session and check login status.

```
session_start();
if (isset($_SESSION["user_logged_in"])) {
   echo "User is logged in.";
} else {
   echo "User is not logged in.";
}
```

---

41. List the common HTTP methods used in web services.

1. **GET**
2. **POST**
3. **PUT**
4. **DELETE**

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